INCH-POUND

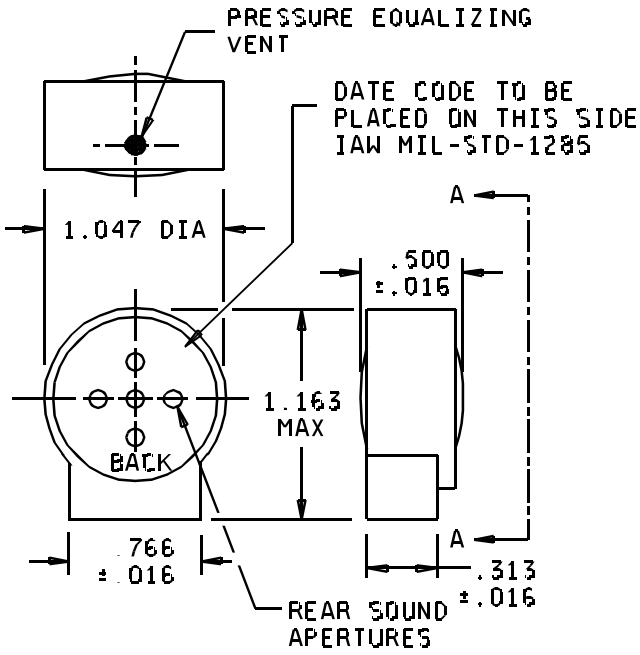
MIL-PRF-26542/4C
30 May 1997
SUPERSEDING
MIL-M-26542/4B
29 March 1985

PERFORMANCE SPECIFICATION SHEET

MICROPHONE ELEMENT, M-101/AIC

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-26542.

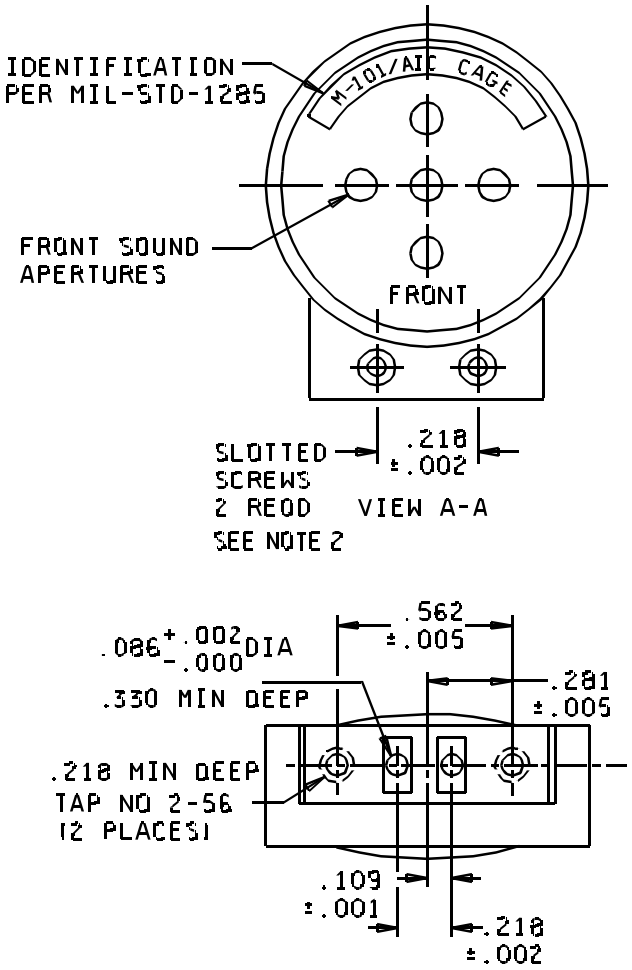


Ⓒ FIGURE 1. Microphone element M-101/AIC.

Ⓒ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:53Z
Check the source to verify that this is the current version before use.



NOTES:

1. Dimensions are in inches. Unless otherwise specified, tolerances are ±.015 (0.38 mm).
2. Set-screws shall hold the part securely to the bracket, shall be slotted (for interchangeability of spares among manufacturers and tri-Service applications), and recessed.
3. Dimensions of microphone are intended to provide interface to a variety of tri-Service brackets, including the US Air Force MT-2189 (M100/AIC), as well as various US Army mask brackets.
4. Hole depths shall be as specified in order to provide a secure bracket connection which does not compromise the mechanical stability of the microphone base.
5. Sound aperture location and configuration are optional, provided that frequency response, environmental and performance requirements are met as specified.

FIGURE 1. Microphone element M-101/AIC - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:53Z
Check the source to verify that this is the current version before use.

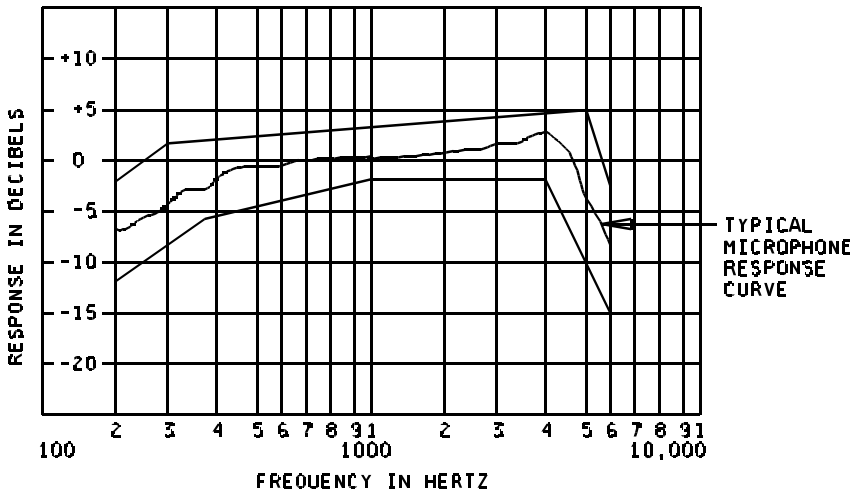


FIGURE 2. Frequency response at sea level.

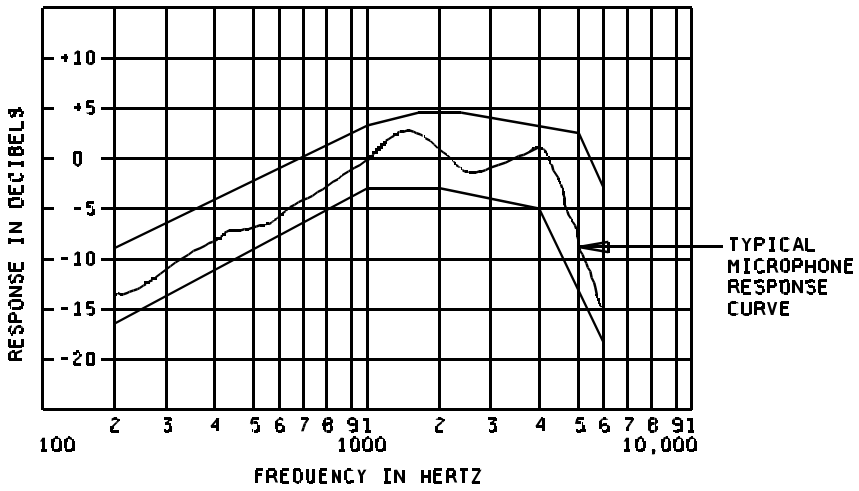


FIGURE 3. Frequency response at 25,000 feet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:53Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Component parts: See figure 1.

Weight: 16 grams, maximum.

Performance:

Sensitivity at ground level: 36.90 dB - 42.92 dB (re 1 µV) or 69.98 µV - 139.63 µV with a SPL input of 28 dynes/cm^2 at 1 kHz.

Sensitivity at a simulated altitude: Sensitivity shall be equal to but not more than 8 dB of the ground level sensitivity when tested at 25,000 feet.

Frequency response at ground level and at 25,000 feet: The envelope shall be as shown on figure 2 and figure 3. The frequency response range of the element shall be 200 Hz to 6,000 Hz.

Ⓒ Impedance: 4.0 ohms to 6.0 ohms. The electrical impedance at any frequency over the range of 200 Hz to 6,000 Hz shall not deviate from the 1,000 Hz impedance by more than 20 percent.

Resistive load: 5.0 ohms.

Intended use: Microphone element M-101/AIC is a noise canceling dynamic moving coil microphone element designed for use on a headband type headset at low altitudes or for use in an oxygen mask or pressure-type oxygen helmet, at altitudes where the use of an oxygen helmet is required. The microphone element is intended for use with intercommunication sets AN/AIC-10( ) and AN/AIC-18 to provide communication under the noise conditions encountered in military aircraft.

The microphone assembly shall be tested in accordance with the tests listed in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:53Z
Check the source to verify that this is the current version before use.

Ⓒ TABLE I. Parameter applicability.

Inspection	Requirement paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group I						
Visual and mechanical inspection	3.1, 3.4, 3.5	4.5.1	X	X		
Sensitivity at ground level	3.5.1	4.5.2.2	X	X		
Sensitivity at altitude	3.5.2	4.5.2.3	X			
Frequency response at ground level	3.5.3	4.5.2.4	X	X		
Frequency response at altitude	3.5.4	4.5.2.5	X			
Impedance	3.5.5	4.5.3	X	X		
Noise cancellation characteristics	3.5.7.1	4.5.5.1	X			
Effect of external magnetic field	3.5.8	4.5.6	X			
Stray magnetic field	3.5.9	4.5.7	X			
Linearity	3.5.10	4.5.8	X			
Talk-out	3.5.11	4.5.9	X	X		
Dielectric withstanding voltage	3.5.12	4.5.10	X			
Signal-to-noise	3.5.7	4.5.5	X		X	
Distortion	3.5.6	4.5.4	X		X	
Interchangeability	3.5.13	4.5.11	X		X	
Group II						
Thermal shock	3.5.14	4.5.12	X			X
Humidity	3.5.15	4.5.13	X			X
Drop	3.5.16	4.5.14	X			X
Pressure equalization	3.5.17	4.5.15	X			X
Explosive decompression	3.5.18	4.5.16	X			X
Salt fog	3.5.19	4.5.17	X			X
Group III						
Vibration	3.5.20	4.5.18	X			X
Bounce	3.5.21	4.5.19	X			X
Altitude	3.5.22	4.5.20	X			X
Moisture barrier seal	3.5.23	4.5.21	X			X
Immersion	3.5.24	4.5.22	N/A			
Group IV						
Fungus	3.5.25	4.5.23	X			
Group V						
Gun blast	3.5.26	4.5.24	N/A			

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:53Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME
Navy - AS, OS
Air Force - 99
DLA - CC

Preparing activity:
Air Force - 85

(Project 5965-0238-04)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:53Z
Check the source to verify that this is the current version before use.